Stabilization of waves in exterior domains with boundary feedback

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Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Wave equation on unbounded domains

Consider $\mathcal{O} \subset \mathbb{R}^n$ a bounded open set with smooth boundary $\partial\mathcal{O}$, where n is an odd number. Denote $\Omega := \mathbb{R}^n \setminus \mathcal{O}$.

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_t^+ \times \Omega, \\ u = -\frac{\partial}{\partial \nu}(G u_t), & \mathbb{R}_t^+ \times \partial\Omega, \\ (u, \partial_t u)(0, x) = (f_1, f_2), & \Omega. \end{cases} \tag{Ψ}$$

Here ν is the exterior normal vector of $\partial\Omega$ and the feedback law is given by

$$G := (-\Delta)^{-1} : \dot{H}^{-1}(\Omega) \to \dot{H}_0^1(\Omega).$$

The energy space considered is $H := L^2(\Omega) \times \dot{H}^{-1}(\Omega)$.
Idea: Prove the exponential stabilization of the local energy

$$E_R(u)(t) = \frac{1}{2}\left[\|u(t)\|^2_{L^2(\Omega_R)} + \|\partial_t u(t)\|^2_{\dot{H}^{-1}(\Omega_R)}\right],$$

where $\overline{\mathcal{O}} \subset B_R$ and $\Omega_R := \Omega \cap B_R$.

Definition (Exterior geometric control condition)

Let $R, T_R > 0$ and $\omega \subset \partial\Omega$. We say that (ω, T_R) satisfies the exterior geometric control condition in B_R if every generalized bicharacteristic ray γ starting from $T_b^*(\mathbb{R}_+ \times \Omega_R)$ verifies one of the next conditions:

1. γ escapes from $\mathbb{R}_+ \times \Omega_R$ before time T_R, or
2. γ hits $\mathbb{R}_+ \times \omega$ between time 0 and T_R in a non-diffractive point.

Theorem (To be proved!)

Let $R, T_R > 0$ such that the pair $(\partial\Omega, T_R)$ verifies the Exterior Geometric Control Condition. Then $\exists C > 0, \alpha > 0$ such that for every $f \in H$ with support contained in B_R, the solution u of (Ψ) with initial data f satisfies

$$E_R(u)(t) \leq Ce^{-\alpha t} E(u)(0).$$

Geometric Control Condition

Definition (GCC)

Let $\omega \subset M$ be an non-empty open subset. Every generalized geodesic of M traveling at speed 1 meets ω in time $t \in (0, T)$.

Theorem

Assume (ω, T) satisfies GCC. Then, there exists $C > 0$ s.t for any $(v_0, v_1) \in H_0^1(M) \times L^2(M) =: \mathcal{E}$ and v solves Dirichlet wave equation on M, we have

$$\|(v_0, v_1)\|_{\mathcal{E}} \leq C \int_0^T \|\mathbf{1}_\omega \partial_t v(t)\|^2_{L^2(M)} dt.$$



C. Bardos, G. Lebeau, J. Rauch (1992)
Sharp sufficient conditions for the observation, control and stabilization of waves from the boundary

.

What do we know on unbounded domains?

$$\begin{cases} \partial_t^2 u - \Delta u + a(x)\partial_t u = 0, & \mathbb{R} \times \Omega, \\ u = 0, & \mathbb{R}_+ \times \Omega, \\ u(0) = f_1, \ \partial_t u(0) = f_2. \end{cases} \tag{E}$$

Exponential stabilization by means of the same microlocal assumption, Lax-Phillips semigroup and microlocal defect measures

Energy space: $H^1(\Omega) \times L^2(\Omega)$, $\omega = \text{supp}(a) \subset \Omega$



L. Aloui, M. Khenissi (2002)
Stabilisation pour l'équation des ondes dans un domaine extérieur

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And with some boundary feedback?

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_+ \times \Omega, \\ \partial_n u + a(x)\partial_t u = 0, & \mathbb{R}_+ \times \partial\Omega, \\ u(0) = f_1,\ \partial_t u(0) = f_2, & \Omega, \end{cases} \qquad (N_a)$$

Same as before, but with more geometrical work and the celebrated Bardos-Lebeau-Rauch Lifting Lemma since we have dynamics on the boundary

Energy space: $H^1(\Omega) \times L^2(\Omega)$, $\omega = \text{supp}(a) \subset \partial\Omega$.



L. Aloui (2002)
Stabilisation Neumann pour l'équation des ondes dans un domaine extérieur

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Pseudodifferential feedback: Bounded domain

$\overline{\Gamma_1} \cap \overline{\Gamma_0} = \emptyset$, $\text{int}(\Gamma_0) \neq \emptyset$, $\Gamma_1 \cup \Gamma_0 = \partial\Omega$. $G := (-\Delta)^{-1} : H^{-1}(\Omega) \to H_0^1(\Omega)$.

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_t^+ \times \Omega, \\ u = -\frac{\partial}{\partial \nu}(G u_t), & \mathbb{R}_t^+ \times \Gamma_0, \\ u = 0, & \mathbb{R}_t^+ \times \Gamma_1, \\ (u, \partial_t u)(0, x) = (f_1, f_2), & \Omega. \end{cases} \tag{1}$$

Exponential stabilization by GCC + Compactness-uniqueness.

Energy space: $L^2(\Omega) \times H^{-1}(\Omega)$.



C. Bardos, L. Halpern, G. Lebeau, J. Rauch, E. Zuazua (1991)
Stabilisation de l'équation des ondes au moyen d'un feedback portant sur la condition aux limites de Dirichlet

.

Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Well-posedness

We denote $-\Delta$ by A, which is a strictly positive self-adjoint operator. In $\dot{H}^{-1}(\Omega)$ we have the following inner product

$$(v_1, v_2)_{\dot{H}^{-1}(\Omega)} := (A^{-1/2} v_1, A^{-1/2} v_2)_{L^2(\Omega)}.$$

Note that by the hhyperbolic energy estimate we have

$$\begin{aligned} \frac{d}{dt} E_R(u)(t) &= \int_{\partial\Omega} \partial_n G(\partial_t u) \left[(-\Delta)^{-1}\right] (\partial_t u) d\sigma \\ &= -\int_{\partial\Omega} \left|\partial_n \left[(-\Delta)^{-1}\right]\right|^2 d\sigma \leq 0, \end{aligned}$$

which justifies the chosen feedback. Hereafter we will use $U_a(t)$ to denote the semigroup associated to

$$\partial \begin{pmatrix} u \\ v \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & I \\ \Delta & 0 \end{pmatrix}.$$

Well-posedness

Consider the unbounded operator A_a which operates as before, with domain

$$D(A_a) = \{(f_1, f_2) \in H : \ (f_2, \Delta f_1) \in H, \ f_1 + \partial_\nu(Gf_2) = 0 \text{ on } \partial\Omega.\}$$

Proposition

A_a is a maximal dissipative operator.

Proof: By the energy estimate, A_a is dissipative. It suffices to prove that $\mathrm{Im}(A_a - I)$ is closed and dense in H which is classical using completeness of the space + Lax-Milgram Theorem, respectively. □

Hence by Hille-Yosida Theorem, A_a generates a contraction semigroup $(U_a(t))_{t\geq 0}$ on H. For $G \equiv 0$, A_a is an anti-adjoint operator and it generates an unitary group $(U(t)_{t\in\mathbb{R}})$.

Lax-Phillips semigroup

We define the spaces of outgoing and incoming data as

$$D_{+,R} = \{f \in H : U(t)f = 0 \text{ over } |x| \leq t + R,\ t \geq 0\},$$
$$D_{-,R} = \{f \in H : U(t)f = 0 \text{ over } |x| \leq -t + R,\ t \geq 0\}.$$

Consider H_0 as the completion of $(C_0^\infty(\mathbb{R}^n))^2$ with the $H^1(\mathbb{R}^n)$ norm and identify H as a subspace. We denote by $D_\pm$ to the previous spaces when $R = 0$. Define the extension operator

$$E : H \to H_0, \quad E = \chi R,$$

where R is the exntesion by reflection and $\chi \in C^\infty(\mathbb{R}^n)$ such that

$$\chi = \begin{cases} 0, & \text{over a compact subset contained in } \mathcal{O}, \\ 1, & \text{on a neighbourhood of } \overline{\Omega}. \end{cases}$$

Then $ED_+ = D_+$ and $ED_- = D_-$.

Lax-Phillips semigroup

For $t \geq 0$, we write

$$Z_a(t) = P^+ U_a(t) P^-,$$

where P^+ (resp. P^-) denotes the projection onto $D_+^\perp$ (resp. $D_-^\perp$). Then $Z_a(t)$ verifies

Proposition

We have

1. $Z_a(t)D_+ = Z_a(t)D_- = 0$.
2. $Z_a(t)$ *operates over* $K = D_+^\perp \cap D_-^\perp$.
3. $(Z_a(t))_{t\geq 0}$ *generates a semigroup over* K.

Proof. 1. By definition $Z_a(t)D_- = 0$. For the other, let $f \in D_+$ and $g \in D_-$. Then

$$(f, g) = (Ef, Eg) = 0,$$

that is $f \in D_-^\perp$ and therefore $P^- f = f$. Since $f \in D_+$, $U_a(t)f = U(t)f \in D_+$, we get that $Z_a(t) = 0$.

Proof of the Proposition

2. Follows from

Lemma

Let B_1, B_2 be two maximal dissipative operators over a Hilbert space H. Suppose that B_1 is an extension of B_2 and $\overline{D(B_2)} = H$. Then $B_1 = B_2$.

$+$ definition of A_a^* and $D(A_a^*)$, which yields

$$0 = (Z_a(t)f, g) = (f, U_a^*(t)g), \quad \forall g \in D_-.$$

3. First we note $Z_a(t)f \in D_-^\perp$ for $f \in D_-^\perp$. Let $f \in K$, then

$$\begin{aligned} Z_a(t)Z_a(s)f &= P^+U_a(t)P^+U_a(s)f = P^+U_a(t)U_a(s)f \\ &= P^+U_a(t+s)f \\ &= Z_a(t+s)f \end{aligned}$$

Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Why Microlocal Analysis?

1. More generality: Accepts different type of operators and regularity on coefficients.
2. **Elliptic world:** Microlocal Carleman Estimates due to Hörmander's "subellipticity" property
3. **Parabolic world:** Spectral inequalities for different operators.
4. **Hyperbolic world:** Propagation of regularity, smallness and compactness.

Microlocal analysis!

Hörmander's idea: Given a Riemannian manifold M, the natural space to do analysis is T^*M instead of M. By Fourier transform,

$$\text{Differential Operators} \quad \subset \quad \text{Pseudodifferential Operators}$$

is equivalent to

$$\text{Homogeneous Polynomials} \quad \subset \quad \text{Symbols}$$

There's a bijection between ΨDOs $\Psi^m(M)$ and symbols $S^m(T^*M)$.

Given $A \in \Psi^m(M)$, the principal symbol is $[\sigma_A^{\text{full}}] \in S^m(T^*M)/S^{m-1}(T^*M)$ and defines a surjective homomorphism

$$\sigma : \Psi^m(M)/\Psi^{m-1}(M) \to S^m(T^*M)/S^{m-1}(T^*M).$$

Manifolds without boundary: Bicharacteristics

Given $P \in \Psi^m(M)$ with principal symbol p singularities live in $p^{-1}(0)$ and are carried by the Hamiltonian field,

$$H_p(x,\xi) = \sum_{j=1}^{n} \frac{\partial p}{\partial \xi_j}\frac{\partial}{\partial x_j} - \frac{\partial p}{\partial x_j}\frac{\partial}{\partial \xi_j}.$$

The bicharacteristic curve of p is a maximal solution $\gamma : I \to T^*M$ such that

$$\dot{x}_j = \frac{\partial p}{\partial \xi_j}, \quad \dot{\xi}_j = -\frac{\partial p}{\partial x_j}, \quad (x(0),\xi(0)) = (x_0,\xi_0).$$

Projection $T^*M \ni (x,\xi) \mapsto x \in M$ are geodesics.

Manifolds without boundary: Wavefront set

Main idea: Study the propagation of some property as regularity (wavefront set) and compactness (microlocal defect measure).
Given $u \in \mathcal{D}'(M)$, we say that $(x_0, \xi_0) \notin \mathrm{WF}(u)$ iff there exists a neighbourhood of $V(x_0)$ and a conical neigh. $W(\xi_0)$ s.t $\varphi \in C_0^\infty(V)$ one has

$$\forall k \in \mathbb{N}, \exists C_k > 0 : |\widehat{\varphi u}(\xi)| \leq C_k(1 + |\xi|)^{-k}, \quad \forall \xi \in W.$$

This notion not only gives the singularity, it also shows the directions where it propagates.

Theorem (Hörmander 1971)

Let $P \in \Psi^m(\mathbb{R}^n)$, classical with real principal symbol. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ s.t $Pu \in C^\infty(\mathbb{R}^n)$ and let γ be a bicharacteristic curve of P. Then

$$\gamma \subset \mathrm{WF}(u), \quad \text{or} \quad \gamma \cap \mathrm{WF}(u) = \emptyset.$$

Manifolds without boundary: Microlocal Defect Measures

Theorem (Gérard 1991)

Let $\{u_n\}_{n\in\mathbb{N}} \subset L^2_{loc}(M)$ weakly converging to 0. Then there exists a subsequence, a Radon measure $\mu \in \mathcal{M}(S^*M)$ such that for every $A \in \Psi^0(\Omega)$ with principal symbol $\sigma \in T^*M$, we have

$$(A(\chi u_n), \chi u_n)_{L^2(\Omega)} \to \int_{S^*M} \sigma(x,\xi) d\mu,$$

for every $\chi \in C_0^\infty(M)$ such that $\chi = 1$ in $\Pi_x(\mathrm{supp}(\sigma))$.

In particular, $\mu = 0$ in S^*M iff $u_n \to 0$

Theorem

Let $P \in \mathrm{Diff}^m(M)$ and $\{u_n\}_{n\in\mathbb{N}} \subset L^2_{loc}(M)$ weakly converging to 0 with MDM μ. TFAE,

1. $Pu_n \to 0$ *in* $H^{-m}_{loc}(\Omega)$,
2. $\mathrm{supp}(\mu) \subset \{(x,\xi) \in S^*M : \sigma_m(P)(x,\xi) = 0\}$.

An example: Bicharacteristics

For the wave operator $\Box = \partial_t^2 - \Delta$, the principal symbol is given by $p(\tau, \xi) = \tau^2 - |\xi|^2$. Bicharacteristics satisfies

$$\dot{t} = 2\tau, \quad \dot{x} = -2\xi, \quad \dot{\tau} = \dot{\xi} = 0,$$

so with the initial condition $(t_0, x_0, \tau_0, \xi_0) \in T^*\mathbb{R}^{1+n}$ such that $\tau_0^2 = |\xi_0|^2$, we get the bicharacteristic curve

$$\gamma(s) = (t_0 + 2s\tau_0, x_0 - 2s\xi_0, \tau_0, \xi_0), \quad s \in \mathbb{R}.$$

The projection $(t, x; \tau, \xi) \to (t, x)$ is called rays of the geometric optics.

An example: Propagation of regularity

Consider $u \in L^2((0,T)\times\Omega)$, $\omega \subset \Omega$ and

$$\begin{cases} \Box u = \partial_t^2 u - \Delta u = 0, & (0,T)\times\Omega, \\ \partial_t u \in L^2((0,T)\times\omega). \end{cases}$$

Then $u \in H^1_{\text{loc}}((0,T)\times\omega)$. Indeed,

$$u \in H^1_{\text{loc}} \iff \mathrm{WF}_1(u) = \emptyset \quad \wedge \quad \partial_t u \in L^2 \iff \mathrm{WF}_0(\partial_t u) = \emptyset.$$

Microlocal regularity: $\mathrm{WF}_{s-m}(Pu) \subset \mathrm{WF}_s(u) \subset \mathrm{Char}(P) \cup \mathrm{WF}_{s-m}(Pu)$.
Therefore intersecting,

$$\mathrm{WF}_1(u) \subset \mathrm{Char}(\Box) \cap \mathrm{Char}(\partial_t) = \emptyset,$$

we get $\mathrm{WF}_1(u) = \emptyset$, which gives $u \in H^1_{loc}((0,T)\times\omega)$.

An example: Why GCC is so important

Suppose that GCC holds and

$$\begin{cases} \Box u = 0, \quad (0,T)\times\Omega, \\ u = 0, \quad (0,T)\times\partial\Omega, \\ u \in L^2((0,T)\times\Omega), \\ \partial_t \in L^2((0,T)\times\omega). \end{cases}$$

By previous example $u \in H^1_{\text{loc}}((0,T)\times\omega)$. Take $\rho_0 = (t_0, x_0; \tau_0, \xi_0) \in (0,\varepsilon)\times\Omega\times\mathbb{R}^{1+n}\setminus\{0\}$, by GCC, the bicharacteristic γ_0 issued from ρ_0 enters $(0,T)\times\omega$, where $u \in H^1((0,T)\times\omega)$. By Hörmander's theorem, $\gamma \subset \mathrm{WF}_1(u)$, so $u \in H^1((0,T)\times\Omega)$

Manifolds with boundary

Denote by $M = \mathbb{R} \times \Omega$ and write $p(t, x, \tau, \xi) = \tau^2 - |\xi|^2$ for the principal symbol of $\Box = \partial_t^2 - \Delta$,

$$\text{Char}(\Box) = \Sigma = \{(t, x, \tau, \xi) \in T^*\overline{M} : p(t, x, \tau, \xi) = 0\},$$

the characteristic variety and $T_b^*M = T^*M \cup T^*\partial M$ the cotangent space at boundary. Consider the following natural projection

$$\pi : T^*\overline{M} \to T_b^*M, \quad \pi(t, x, \tau, \xi) = \begin{cases} (t, x, \tau, \xi), & (t, x) \in M \\ (t, x, (\tau, \xi)\,|_{T_{(t,x)}\partial M}), & (t, x) \in \partial M. \end{cases}$$

Manifolds with boundary: Generalized bicharacteristics

Definition

We define the subsets $\mathcal{E}, \mathcal{G}$ and $\mathcal{H}$ of $T^*\partial M$ as

$$\mathcal{E} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 0\},$$
$$\mathcal{G} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 1\},$$
$$\mathcal{H} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 2\}$$

$\mathcal{E}, \mathcal{G}$ and $\mathcal{G}$ are, respectively, the set of elliptic, glancing and hyperbolic points.

- Let $\rho \in \mathcal{G}$, ρ is said strictly diffractive if and only if $H_p^2(y_n)(\rho) > 0$. The set of strictly diffractive points is denoted by $\mathcal{G}_d$.
- Let $q \in T^*\partial M$, q is said a non diffractive point if and only if $q \in \mathcal{E} \cup \mathcal{H}$ or $q \in \mathcal{G}$ and $\forall \varepsilon > 0$, there exists $s \in (-\varepsilon, \varepsilon) \setminus \{0\}$ such that $\gamma(s) \notin T^*\mathbb{R}^{n+1}\,|_{\overline{M}}$ where γ is the bicharacteristic curve of $\Box$ starting at a point $\tilde{q} = \pi^{-1}(q) \cap \Sigma$.

Boundary interaction

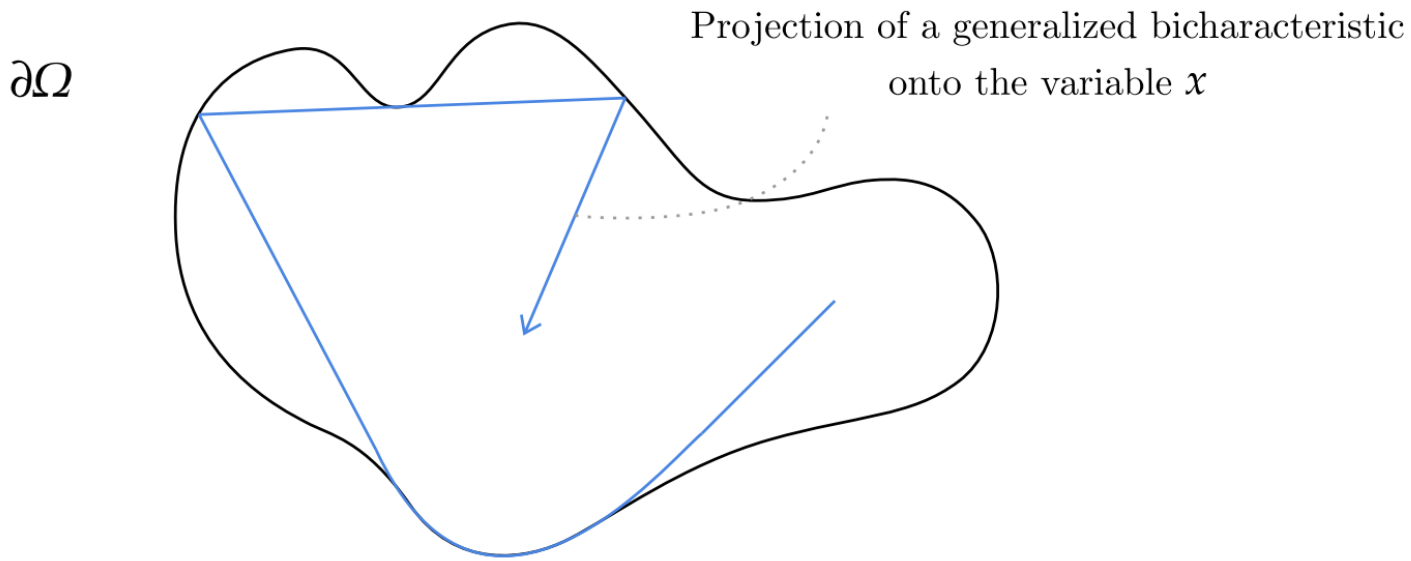


Figure 1.1: Generalized bicharacteristics

C. Loyola (2025)
Stabilisation, contrôle et prolongement unique pour quelques équations aux dérivées partielles nonlinéaires dispersives (Thèse de Doctorat en Sciences)

Manifolds with boundary: propagation results

Theorem (Melrose-Sjöstrand)

Let u be a distribution over M, extendable such that $\Box u \in C^\infty(\overline{M})$ and over ∂M we have $u = 0$ or $\partial_n u = 0$. If γ is a generalized bicharacteristic ray of $\Box$, then we have:

1. $\mathrm{WF}_b(u) \subset \Sigma_b = \pi(\Sigma)$,
2. $\gamma \subset \mathrm{WF}_b(u)$ *or* $\gamma \cap \mathrm{WF}_b(u) = \emptyset$.

Theorem (Lebeau)

Denote by $G(s)$ the generalized bicharacteristic flow and by μ the MDM. Suppose that ∂M doesn't have a contact point of infinite order. Then there exists $C > 0$ such that for all $s \geq 0$, $\mu^+(G(s)B) \leq C\mu^+(B)$ for every Borelian B on $\pi(\Sigma)/\mathbb{R}^*_+$.

Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Lifting lemma

Theorem (Lifting lemma, Neumann case)

Let $q = (t_0, x_0'; \tau_0, \xi_0') \in T^*\partial M$ be a non-diffractive point and $I \times \omega$ a neigh. of $(t_0, x_0 = (x_0', 0))$ in M. Let $\{u_k\}_{k\in\mathbb{N}}$ be a solution of (N_a) verifying $u_k \rightharpoonup 0$ in H^1_{loc} and μ the associated MDM. Suppose that

$$u_k \mid_{\partial M} \to 0 \text{ in } H^1(I \times \partial\omega) \text{ and } \partial_n u_k \mid_{\partial M} \to 0 \text{ on } L^2(I \times \partial\omega).$$

Then $\mu \equiv 0$ around q.

In our case: Main problem is to replicate ideas with less regularity assumptions.

Stabilization result: Aloui's Neumann case

It's not hard to prove that $U_a(t)f = Z_a(t)f$ over B_R. It suffices to prove $\|Z_a(T)\| < 1$ for some $T > 0$.

Lemma

For $T = T_R + 9R$, $\|Z_a(T)\| < 1$.

"Proof": By contradiction, $\exists\{f_k\}_{k\in\mathbb{N}} \subset K : \|f_k\| = 1$ and $\|Z_a(T)f_k\| \to 1$. Since $\|U_a(t)f_k\| \leq \|f_k\|$,

$$\Pi_x(U_a(t)f_k) := u_k \text{ is bounded in } H^1_{\text{loc}} \text{ with } \mu \text{ as MDM}$$

then we need

Lemma

$\mu \equiv 0$ *on* $T^*(K(T'))$ *where* $T' = T_R + 3R$ *and*

$$K(T') = \{(t,x) \in \mathbb{R}_+ \times \Omega : |x| \leq t - T' + R,\ t \geq T'\}.$$

Lemma's proof: using EGCC

Let γ the bichar. with $q \in T^*(K(T'))$ as initial point.

Case 1: γ does not pass $\partial\Omega$ or it hits $\partial\Omega$ after time $t_0 > 2R$

Denote by μ_0 the MDM attached to $\Pi_x(U(t)f_k)$. We have $\mu = \mu_0$ on $\{\gamma(t) : 0 \leq t \leq t_0\}$. By Lax-Phillips semigroup, $\mu_0 = 0$ around $\gamma(t_1)$, where $t_1 < t_0$ is s.t $\gamma(t_1) \in T^*(K(2R))$. By propagation theorem, $\mu = 0$ around q.

Case 2: γ hits $\partial\Omega$ for $t_0 \leq 2R$.

Note that $\gamma(T'), \gamma(t_0) \in B_R$ and $T' - t_0 > T_R$. By EGCC, γ passes through $\omega \times [t_0, T]$ on a non-diffractive point $\tilde{q}$. By energy estimates,

$$(u_k \mid_{\partial M}, \partial_n u_k \mid_{\partial M}) \to (0,0) \; H^1(\omega \times [t_0, T]) \times L^2(\omega \times [t_0, T]).$$

By Lifting Lemma, $\mu = 0$ around $\tilde{q}$, and by propagation theorem $\mu = 0$ around q. □

Back to previous lemma

The first part comes from the following domain of dependence result

Lemma

$\forall g \in D_{-}^{\perp}$, $\|Z_a(2R)g\| \leq 2\|g\|_{5R}$.

and it implies $\|Z_a(T)f\| = \|f\| = 1$. Then by

Lemma

$G_T = \{f \in K : \|Z_a(T)f\| = \|f\|\} = \{0\}$

we achieve a contradiction.

Sketch of the proof: We need to prove 1. $\dim(G_T) < +\infty$, 2. A operates over G_T and 3. A doesn't have eigenvalues. Hence $f \in K$ satisfies an elliptic equation with analytics coefficients. Hence f is analytic, and by Dirichlet boundary condition, $f \equiv 0$.

Contents

1. Motivation and some previous works
2. Well-posedness and Lax-Phillips semigroup
3. An introduction to Microlocal Analysis
4. Lifting lemma and Stabilization result
5. Future work

Some extensions

1. EGCC forces the damping mechanism. It is possible to have mixed boundary conditions? (i.e, $u = 0$ on $\Gamma_1 \subset \partial\Omega$)
2. There's an extension in the Dirichlet case with a decreasing non-linearity with polynomial growth f and $a \in L^\infty_+(\Omega)$ with compact support,

$$\begin{cases} \partial_t^2 u - \Delta u + a(x) f(\partial_t t u) = 0, & \mathbb{R}^+_t \times \Omega, \\ u = 0, & \mathbb{R}^+_t \times \partial\Omega, \\ (u, \partial_t u)(0, x) = (\varphi_1, \varphi_2), & \Omega. \end{cases}$$

Is it possible to extend to the Neumann and Pseudodifferential case?



M Daoulatli (2007)
Local energy decay for the nonlinear dissipative wave equation in an exterior domain

Thanks for your attention!